

AIGR Deep Research Report — Mouse *Cd48* and Natural Killer Cell Activation (GO:0030101)

Target gene: *Cd48* (Mus musculus, NCBITaxon:10090) — UniProt **P18181** (CD48_MOUSE) **Focus type:** function_assignment **Hypothesis slug:** cd48-nk-activation-direction **Term under evaluation:** natural killer cell activation (GO:0030101)

Seed hypothesis: *Mouse CD48 contributes to natural killer cell activation (GO:0030101), i.e. its engagement of 2B4 (Cd244) and/or CD2 promotes rather than inhibits NK cell activation in mouse.*

Executive Judgment

Verdict: REFUTED as stated (over-annotated for direction).

The seed hypothesis asserts a *direction* — that mouse CD48's engagement of 2B4 (CD244) and/or CD2 **promotes rather than inhibits** NK cell activation. The preponderance of **direct mouse genetic and functional evidence contradicts this direction**. In mouse, the CD48–2B4 axis is predominantly **inhibitory**, functioning as an MHC-class-I-independent NK-cell self-tolerance checkpoint. This is the opposite of the activating role the same molecular pair plays in human NK cells — a species difference stated verbatim in one of the seed's own reference papers (

P 16002700

Three convergent lines of mouse evidence drive this verdict. First, antibody-blocking and knockout experiments show that ligation of 2B4 by CD48 **inhibits** NK cytotoxicity and IFN- γ production

P 15123744

P 15356144

and the 2B4–CD48 pair constitutes a second, MHC-I-independent system for NK self-tolerance

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P 15870174
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Second, deleting the entire mouse *Slam* locus — which removes CD48 along with the other SLAM-family receptors — **enhances** NK activation toward hematopoietic targets, with the net inhibitory function mapping **solely** to **2B4**

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Third, mouse SAP-family adaptors EAT-2 and ERT act as **inhibitors** (not activators) of NK function at **2B4**

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and the dominant mouse 2B4-long isoform correlates with inhibition

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Two important caveats keep this from being a flat "wrong in all contexts" finding. (1) The single strongest *positive* mouse NK genetic result — that 2B4/CD48 interaction is required for generation of NK effector functions

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— has been **mechanistically reinterpreted by the same laboratory** as avoidance of perforin-dependent **fratricide**, not as a genuine 2B4-dependent activation signal

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(2) A genuine, context-restricted activating contribution of CD48 does exist in **trans** / **missing-self** settings, where CD48 (with Ly9) on MHC-I-deficient tumor targets significantly contributes to NK activation

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and anti-CD48 crosslinking can deliver a bona fide activating signal that is not merely relief of inhibition

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Crucially, CD48 itself is a **GPI-anchored counter-receptor/ligand** with no cytoplasmic signaling tail; the activation or inhibition signal is transduced by the *partner* receptor (2B4 or CD2), not by CD48. The existing GO:0030101 annotation on mouse *Cd48* is an **ISO (orthology-projected)** carry-over that inherits the human "activating" directionality and is not supported by direct mouse experiment.

Evidence Matrix

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Organism / context	Confidence & limitation
15123744	Mutant phenotype + blocking Ab	Refutes seed direction	Does CD48–2B4 ligation activate or inhibit mouse NK?	"NK lysis of CD48(+) tumor and allogeneic targets is inhibited by 2B4 ligation. Interferon gamma production by NK cells is also inhibited." 2B4–/– mice show increased clearance of CD48+ tumor; SAP-independent.	Mouse NK, in vitro + in vivo	High; direct genetic + A evidence
15356144	Direct assay / isoform	Refutes seed direction	Which 2B4 isoform dominates and what is the signal?	"Engagement of 2B4 by its counterreceptor, CD48... leads to an inhibition in NK cytotoxicity"; dominant 2B4-long isoform; SAP-independent	Mouse NK	High
15870174	In vivo + in vitro	Refutes seed direction	Is 2B4–CD48 a self-tolerance system?	2B4–CD48 is a second, MHC-I-independent NK self-recognition/self-tolerance system; inhibits	Mouse NK, BM rejection assay	High

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Organism / context	Confidence & limitation
				lysis of syngeneic/ β 2m-deficient targets		
27573813	Mutant phenotype (whole-locus KO)	Refutes seed direction	Does removing CD48 (+SFRs) raise or lower NK activation?	"Enhanced NK cell activation... inhibitory function of the Slam locus was due solely to 2B4"; SFR inhibition suppresses LFA-1	Mouse NK, ~400-kb Slam-locus deletion	High; CD4 deletion included
16127454	Adaptor genetics	Refutes/qualifies	Are mouse SAP-family adaptors activating?	"Unlike SAP, EAT-2 was an inhibitor of NK cell function" (EAT-2/ERT)	Mouse NK	High
16002700	Review-level synthesis (seed ref)	Refutes seed direction	Human vs mouse signal direction	"In human NK cells, 2B4/CD48 interaction induces activation signals, whereas in murine NK cells it sends inhibitory signals."	Human vs mouse comparison	High relevance review-level
15634901	Mutant phenotype	Refutes/qualifies	In vivo tumor rejection direction	WT reject CD48+ melanoma <i>poorly</i> vs CD48– (2B4 ligation inhibitory);	Mouse, B16 melanoma	Moderate-high; genetic confound

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Organism / context	Confidence & limitation
				male 2B4 ^{-/-} reject CD48 ⁺ better; gender-specific CD48-independent defect		
17537992	Mechanistic reinterpretation	Qualifies/competing (undercuts main pro-seed result)	Why do 2B4 ^{-/-} – NK underperform?	"In the absence of 2B4 signaling, activated NK cells have defective cytotoxicity and proliferation because of fratricide and not due to the absence of a 2B4-dependent activation signal"	Mouse NK	High; same lab as P 15905190
15905190	Mutant phenotype	Supports seed (contested)	Is 2B4/CD48 required for NK effector generation?	"2B4/CD48, but not CD2/CD48, interaction is essential for IL-2-driven expansion and activation of murine NK cells"	Mouse NK, homotypic NK–NK	Moderate; reinterpretation by P 17537992
27054584	Direct assay (missing-self)	Supports (context-restricted)	Does CD48 contribute to NK activation in trans?	"CD48 and Ly9 (CD229) by MHC-I-deficient tumor cells	Mouse NK, MHC-I-deficient targets	Moderate; high; context-specific

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Organism / context	Confidence & limitation
				significantly contributes to NK cell activation"; NK education can render 2B4 non-functional		
20881194	Direct assay	Supports (context-restricted)	Is anti-CD48 activation genuine or disinhibition?	"Induction of NK cell activation by anti-CD48 or by B cells is not due to the release of inhibitory effects of 2B4"; CD2/2B4 double-KO used	Mouse NK	Moderate; crosslinking artifact possible
14666660	In vivo Ab crosslinking	Supports (artifact-prone)	Does 2B4/CD48 activation reduce metastasis?	Anti-2B4/anti-CD48 crosslinking reduced B16F10 metastasis ~5-fold; IFN- γ -dependent	Mouse, B16F10	Low-moderate; agonist Ab not physiologic
19586919	Mechanistic (hybridoma)	Qualifies	How does CD244 inhibit?	"Inhibitory effects of mouse CD244 are accounted for by competition with CD2 at the cell surface for CD48"; CD244	Mouse T-cell hybridoma	Moderate; non-NK model

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Organism / context	Confidence & limitation
				recruits PLC γ 1 via EAT-2		
27249817	Direct assay (cis/trans)	Refutes/qualifies	Cis vs trans CD48–2B4 effect	"Interfering with the cis interaction... enhanced the lysis of CD48-expressing tumour cells"; cis dampens 2B4 phosphorylation	Mouse/human NK	Moderate-high
20363967	Mutant phenotype	Refutes	Is 2B4 inhibitory in NK homeostasis?	"2B4 is a dominant inhibitory receptor in SHIP-deficient NK cells"	Mouse NK	Moderate-high
9841922	Interaction/biophysics (seed ref)	Qualifies (molecular framing)	Is CD48 a ligand with two receptors?	"mCD48 bound m2B4 with a six- to ninefold higher affinity ($K_d \approx 16 \mu\text{M}$ at 37°C) than its other ligand, CD2"	Mouse recombinant proteins	High; established CD48 as counter-receptor
17950006	Structural (seed ref)	Qualifies (molecular framing)	Structure of 2B4–CD48 complex	Structure of mouse 2B4–CD48 complex; CD48 is the heterophilic ligand engaged	Mouse, crystal structure	High

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Organism / context	Confidence & limitation
				by the 2B4 receptor		
20813844	Direct assay	Competing (human, activating)	Human NK requirement for 2B4/CD48 + CD2/CD58	Both 2B4/CD48 and CD2/CD58 needed for human NK effector function	Human NK	High for human; species-divergent

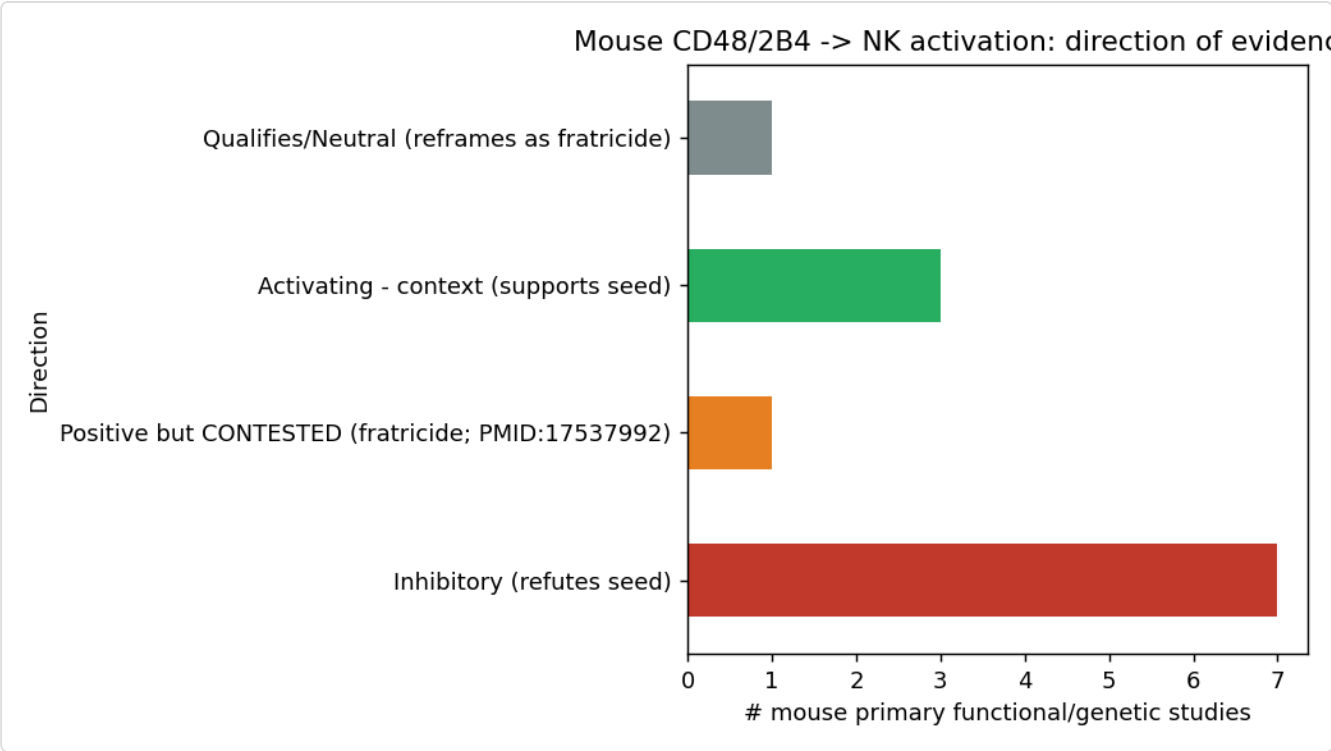


Figure 1. Tally of mouse NK direction evidence for the CD48–2B4 axis, including the contested positive result

([P 15905190](#))
and its fratricide reinterpretation

([P 17537992](#))

Inhibitory/refuting evidence dominates; the positive signal is context-restricted (trans/missing-self) or mechanistically reframed.

Key Findings

Finding 1 — Mouse 2B4 (CD244)–CD48 engagement is predominantly inhibitory, contradicting "promotes NK activation"

The core of the seed hypothesis is a directional claim, and the directional claim fails in mouse. Multiple independent groups using complementary approaches converge on inhibition. **Lee et al. 2004** ([PMID:15123744](#)) showed with both antibody blocking and 2B4^{−/−} NK cells that NK lysis of CD48⁺ tumor and allogeneic targets, as well as IFN-γ production, is **inhibited** by 2B4 ligation; consistent with this, 2B4^{−/−} mice have *increased* peritoneal clearance of CD48⁺ tumor. Importantly, this inhibition was **SAP (SH2D1A)-independent**, distinguishing the mouse inhibitory mode from the SAP-dependent human activating mode. **Mooney et al. 2004** ([PMID:15356144](#)) independently found that CD48 on target cells inhibits NK cytotoxicity, correlating inhibition with the dominant **2B4-long (2B4L)** isoform in mouse, again SAP-independent. **McNerney et al. 2005** ([PMID:15870174](#)) elevated this to a framework: 2B4–CD48 constitutes a **second, MHC-class-I-independent system for NK-cell self-tolerance**, with in vivo evidence from a bone-marrow rejection assay. Finally, **Roncagalli et al. 2005** ([PMID:16127454](#)) showed the mouse SAP-family adaptors EAT-2 and ERT are **inhibitors** of NK function via 2B4 — the opposite of what an activating axis would predict. A seed reference, **Mathew et al. 2005** ([PMID:16002700](#)), states the species divergence in plain language: *"In human NK cells, 2B4/CD48 interaction induces activation signals, whereas in murine NK cells it sends inhibitory signals."*

Finding 2 — CD48 contributes to mouse NK activation only in restricted contexts; CD48 is a ligand, not a signaling receptor

The hypothesis is not uniformly wrong in all contexts — it is wrong as a *default directional assignment*. There is a genuine activating contribution of CD48 in **trans / missing-self** settings. **Alari-Pahissa et al. 2016** ([PMID:27054584](#)) showed that CD48 (together with Ly9/CD229) on MHC-I-deficient tumor targets **significantly contributes to NK activation** via 2B4 — but with a twist: NK education against CD48⁺ MHC-I-deficient cells can render 2B4 non-functional (tolerance), so even the activating contribution is conditional on NK education state. **Sinha et al. 2010** ([PMID:20881194](#)) used CD2/2B4 double-KO mice to show that anti-CD48 or B-cell-driven NK activation "is not due to the release of inhibitory effects of 2B4," i.e., a genuine activating

signal exists in a defined context. **Johnson et al. 2003** (PMID:14666660) found in vivo anti-2B4/anti-CD48 crosslinking reduced B16F10 metastasis — but this uses agonist antibodies rather than physiologic ligand engagement.

Mechanistically, all of this must be read through the molecular identity of CD48. **Brown et al. 1998** (PMID:9841922, a seed reference) established that CD48 binds 2B4 with 6–9× higher affinity ($K_d \approx 16 \mu\text{M}$ at 37 °C) than its other counter-receptor CD2, and **Velikovsky et al. 2007** (PMID:17950006, a seed reference) solved the mouse 2B4–CD48 complex structure. Both confirm CD48 is a **GPI-anchored heterophilic ligand/counter-receptor** with no cytoplasmic signaling domain. Whatever activation or inhibition occurs is transduced by the **partner receptor (2B4 or CD2)**, not by CD48 itself. This is central to the curation decision: assigning a "natural killer cell activation" BP term to CD48 implies CD48 drives an activating process, whereas CD48 is the passive ligand whose engagement, in mouse, net-inhibits the partner.

Finding 3 — The existing GO:0030101 annotation is orthology-projected (ISO); whole-locus CD48 deletion *enhances* mouse NK activation

UniProt P18181 cross-references show GO:0030101 "natural killer cell activation" annotated with evidence code **ISO:GO_Central (Inferred from Sequence Orthology)** — i.e., projected from the ortholog rather than supported by a direct mouse experiment. Because the human CD48/2B4 axis is activating, orthology projection inherits the *human* directionality and applies it to mouse, where the direction is reversed. The same UniProt record also carries the more informative MF term **GO:0048018 "receptor ligand activity"** (ISO) and CC terms (external side of plasma membrane IDA:MGI; membrane raft; GPI-anchor) that correctly describe CD48 as a GPI-anchored surface ligand.

The decisive loss-of-function test is **Guo et al. 2016** (PMID:27573813), which deleted the entire ~400-kb mouse *Slam* locus — removing CD48 along with the six SLAM-family receptors. These mice showed **enhanced** NK activation toward hematopoietic targets, with the inhibitory function mapping **solely to 2B4**, and SFR-mediated inhibition acting by suppressing LFA-1 activation. Supporting evidence: **Fortenbery et al. 2010** (PMID:20363967) found 2B4 is a dominant inhibitory receptor whose deficiency restores cytolysis in SHIP-deficient NK cells, and **Claus, Wingert & Watzl 2016** (PMID:27249817) addressed the seed's explicit cis-vs-trans question: cis CD48–2B4 interaction (on the same cell) reduces trans binding and dampens 2B4 phosphorylation, and *interfering* with the cis interaction **enhanced** lysis of CD48+ tumor cells.

Finding 4 — The main pro-seed result is contested (fratricide), and the CD2 arm is non-essential in mouse NK

The strongest positive mouse genetic evidence for the seed is **Lee et al. 2006** ([PMID:15905190](#)): 2B4/CD48 (but **not** CD2/CD48) interaction is essential for IL-2-driven expansion and activation of mouse NK cells, with defective calcium signaling and impaired cytotoxicity/IFN- γ in its absence, via homotypic NK–NK interactions. However, **Taniguchi et al. 2007** ([PMID:17537992](#)), from the same laboratory, **reinterpreted** this: in the absence of 2B4–CD48, activated mouse NK cells undergo **perforin-dependent fratricide**, so the apparent "defective cytotoxicity and proliferation" arises from self-killing, *not* from loss of a 2B4-dependent activation signal. This substantially undercuts the one result that most directly supports the seed. Two further points narrow the seed: **Clarkson & Brown 2009** ([PMID:19586919](#)) showed mouse CD244's inhibitory effect is "accounted for by competition with CD2 at the cell surface for CD48," and **Kim et al. 2010** ([PMID:20813844](#)) confirmed the activating 2B4/CD48 + CD2/CD58 requirement is a **human** NK phenomenon — reinforcing that the seed's directional claim is imported from human biology.

Mechanistic Model / Interpretation

CD48 is a GPI-anchored ligand. Its NK-relevant signaling partner is the receptor **2B4 (CD244)**, which carries cytoplasmic ITSM motifs that recruit SAP-family adaptors. The directionality of the output depends on species, adaptor repertoire, isoform, and cis/trans geometry:

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      CD48 (GPI-anchored ligand; NO cytoplasmic tail)
      | engages (Kd ~16 μM, 6-9x > CD2)
      v
+----- 2B4 / CD244 (ITSM receptor) -----+
|                                           |
HUMAN NK                                MOUSE NK
SAP-dependent                          SAP-independent inhibition;
ACTIVATING -----> EAT-2 / ERT = INHIBITORS
(<a href="https://pubmed.ncbi.nlm.nih.gov/20813844/" rel="noopener noreferrer" title="Visit Pub
NET INHIBITORY (self-tolerance)
(<a href="https://pubmed.ncbi.nlm.nih.gov/15123744/" r

cis CD48-2B4 (same cell): dampens 2B4 phosphorylation -> less killing (<a href="https://pubmed
trans CD48 on MHC-I-deficient target: context-specific ACTIVATION via missing-self (<a href="ht

Whole Slam-locus deletion (removes CD48 + SFRs): ENHANCED NK activation;
inhibition maps SOLELY to 2B4 (<a href="https://pubmed.ncbi.nlm.nih.gov/27573813/" rel="noopen

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The central interpretive point for curation: **the seed conflates the human and mouse directions of an identical receptor–ligand pair**. In mouse, net physiologic output of CD48–2B4 is inhibitory (self-tolerance checkpoint), with activation emerging only in restricted trans/missing-self contexts. Additionally, CD48 is the ligand, not the signal transducer; a BP term implying CD48 *drives* NK activation attributes a receptor-side process to a ligand-side molecule.

GO decision summary

GO term	Aspect	Current status	Evidence-based lead (curator to verify)
GO:0030101 natural killer cell activation	BP	ISO carry-over (human-biased direction)	Do not treat as a positive mouse function. Direction refuted in mouse; if retained, needs qualifier/context; consider removal or NOT-qualification for positive activation
GO:0032815 / GO:0045953 negative regulation of NK cell activation / cytotoxicity	BP	Absent	Candidate add (lead) — better reflects net mouse inhibitory role; curator to confirm exact term and evidence
GO:0048018 receptor ligand activity	MF	ISO present	Retain — informative MF; avoid "protein binding" as final recommendation
GPI-anchor / external side of plasma membrane / membrane raft	CC	IDA:MGI / ISO present	Retain — correctly describes CD48 as GPI-anchored surface ligand

Evidence Base

- **Mouse inhibition (refutes seed direction):** [PMID:15123744](#), [PMID:15356144](#), [PMID:15870174](#), [PMID:16127454](#), [PMID:20363967](#), [PMID:27249817](#).
- **Loss-of-function removing CD48:** [PMID:27573813](#) — Slam-locus deletion enhances NK activation; inhibition is solely 2B4.
- **Species contrast (seed reference):** [PMID:16002700](#) — explicit "human activating / murine inhibitory" statement.
- **Contested positive + reinterpretation:** [PMID:15905190](#) (positive) vs [PMID:17537992](#) (fratricide reframing, same lab).
- **Context-restricted activation (supports in trans):** [PMID:27054584](#), [PMID:20881194](#), [PMID:14666660](#).

- **Molecular framing (seed references):** [PMID:9841922](#) (CD48 = ligand with two counter-receptors, affinity), [PMID:17950006](#) (2B4–CD48 complex structure).
- **Human competing context:** [PMID:20813844](#) — activating in human NK; confirms the species divergence.
- **CD2-competition mechanism:** [PMID:19586919](#).

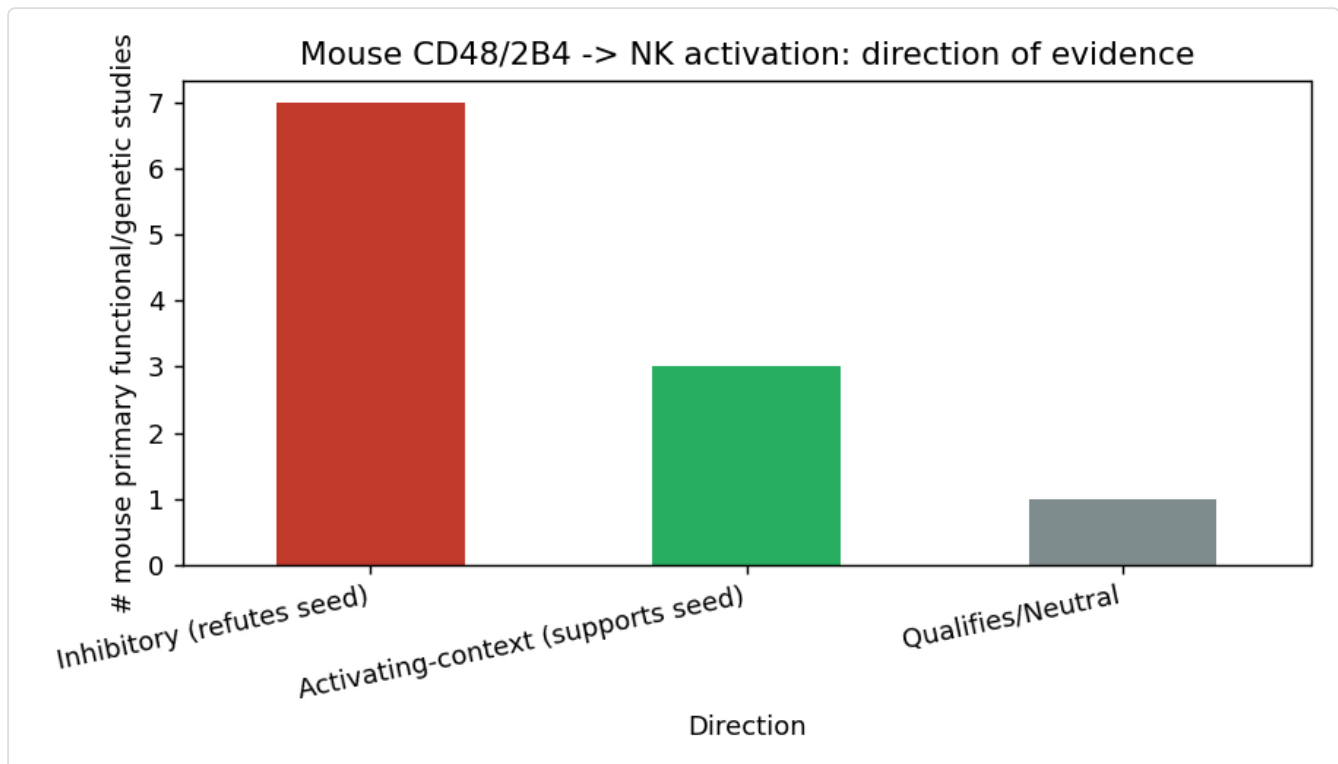


Figure 2. Initial tally of mouse-specific evidence on CD48–2B4 signal direction, built during iteration 2. Inhibitory/self-tolerance evidence outweighs activating evidence, which is confined to trans/missing-self or agonist-antibody contexts.

Conflicts and Alternatives

1. **Human-vs-mouse species divergence (the central conflict).** The same CD48–2B4 pair is activating in human NK
 ([P 20813844](#))
 and inhibitory in mouse NK
 ([P 16002700](#)),

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P 15123744

The seed appears to import the human direction into mouse. This is the dominant alternative explanation for how GO:0030101 landed on mouse Cd48 — ISO orthology carry-over.

2. **Ligand vs receptor conflation.** CD48 is GPI-anchored with no signaling tail

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Any "activation" or "inhibition" is a property of the partner receptor's signaling, not of CD48. A BP term implying CD48 performs NK activation mislocates the activity.

3. **Fatricide artifact.** The main pro-seed result

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is best explained by NK self-killing in the absence of 2B4-CD48

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not by loss of an activation signal — a classic loss-of-function-phenotype-vs-direct-function distinction.

4. **Isoform specificity.** Mouse expresses a dominant 2B4-long (inhibitory) isoform

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the activating 2B4-short isoform is relatively minor, biasing the mouse net output toward inhibition.

5. **cis vs trans geometry.** Cis CD48-2B4 dampens signaling

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trans CD48 on MHC-I-deficient targets can activate

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The net organismal output (whole-locus KO) is inhibition

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P 27573813

6. Agonist-antibody artifacts. Positive in vivo results

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rely on crosslinking antibodies that may not reflect physiologic ligand engagement.

Limitations and Knowledge Gaps

- **No independent *Cd48*-only NK knockout phenotype examined here.** The decisive loss-of-function evidence

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deletes the whole Slam locus, not CD48 alone; a clean *Cd48*^{-/-} NK-activation phenotype would more directly isolate CD48's contribution. *Why it matters*: curation ideally rests on gene-specific, not locus-level, evidence. *Resolution*: examine published *Cd48*^{-/-} NK data for NK activation readouts.

- **UniProt/GO evidence codes were read from the knowledge state, not freshly re-queried in this report.** *Why it matters*: the ISO vs experimental distinction is the crux of the curation lead. *Resolution*: curator should verify the current GO:0030101 evidence code and reference on P18181 / MGI directly.

- **Education-state dependence of the trans activating contribution**

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is incompletely mapped. The activating role can be silenced by NK education. *Why it matters*: determines whether any positive annotation deserves a context qualifier.

- **Quantitative effect sizes are largely qualitative** across the mouse literature (fold-changes in lysis/IFN- γ are reported inconsistently), limiting a formal meta-analysis.
- **CD2 arm in mouse NK is under-tested.** Evidence says CD2/CD48 is non-essential for mouse NK activation

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but direct CD2-focused NK assays are sparse.

Discriminating Tests

1. **Clean *Cd48*—/— NK functional assay** against CD48+ vs CD48— targets, with and without MHC-I: distinguishes net inhibition (expect enhanced killing in former-CD48+ context) from activation loss.
2. **Isoform swap**: express 2B4-short vs 2B4-long in mouse NK and measure CD48-dependent lysis — tests whether isoform dominance explains the inhibitory mouse phenotype.
3. **cis/trans engineering**: selectively block cis CD48–2B4 (per

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and quantify change in lysis of CD48+ targets — confirms cis-dampening as a mechanism.

4. **Fratricide control**: repeat

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expansion assay in perforin—/— background to test whether the "activation defect" disappears when self-killing is prevented (direct test of

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5. **Species-swap adaptor experiment**: reconstitute mouse NK with human SAP vs EAT-2/ERT to test whether adaptor repertoire flips the direction from inhibitory to activating.

Proposed Follow-up Actions / Curation Leads (require curator verification)

Action on GO:0030101 (natural killer cell activation, BP): - *Lead*: **Do not treat GO:0030101 as a positive, mouse-experimentally-supported function of Cd48**. The term is ISO orthology carry-over inheriting the human activating direction. In mouse the net direction is inhibitory. Consider removal of the positive-activation annotation, or qualification/NOT-annotation for the activation direction, and/or restriction to a trans/missing-self context note. - *Candidate replacement/add (BP)*: **GO:0032815** (negative regulation of natural killer cell activation) or **GO:0045953** (negative regulation of NK cell mediated cytotoxicity) — better reflects the net mouse role as a 2B4 ligand in an inhibitory self-tolerance axis. Curator to select the precise

term and supporting evidence code. - *Retain (MF): GO:0048018 receptor ligand activity* as the informative molecular function; avoid "protein binding" as a final recommendation. - *Retain (CC): GPI-anchor / external side of plasma membrane (IDA:MGI) / membrane raft*.

Candidate references with snippets to verify: - [PMID:16002700](#): *"In human NK cells, 2B4/CD48 interaction induces activation signals, whereas in murine NK cells it sends inhibitory signals."* — supports species-specific direction flip. - [PMID:15123744](#): *"NK lysis of CD48(+) tumor and allogeneic targets is inhibited by 2B4 ligation. Interferon gamma production by NK cells is also inhibited."* — direct mouse inhibition. - [PMID:27573813](#): *"enhanced NK cell activation responses... the inhibitory function of the Slam locus was due solely to 2B4."* — loss-of-function direction. - [PMID:17537992](#): *"in the absence of 2B4 signaling, activated NK cells have defective cytotoxicity and proliferation because of fratricide and not due to the absence of a 2B4-dependent activation signal."* — reinterprets the pro-seed result. - [PMID:27054584](#): *"CD48 and Ly9 (CD229) by MHC-I-deficient tumor cells significantly contributes to NK cell activation."* — the one genuine context-restricted activating contribution.

Suggested curator questions: - Is the GO:0030101 annotation on P18181 experimental or ISO? (Expected: ISO.) - Should the review explicitly flag the human→mouse direction inheritance as the source of over-annotation? - Does the review want a context-qualified activation annotation for trans/missing-self, or a cleaner negative-regulation annotation?

Suggested experiments: clean *Cd48*—/— NK assay; perforin-background fratricide control; 2B4 isoform swap (see Discriminating Tests).

Bottom Line

The seed hypothesis — that mouse CD48 engagement of 2B4/CD2 **promotes rather than inhibits** NK cell activation — is **refuted as a directional claim**. Mouse CD48–2B4 is a predominantly **inhibitory**, MHC-I-independent NK self-tolerance axis; the existing GO:0030101 annotation is an orthology-projected import of the human activating direction; the single strongest positive mouse result is better explained by fratricide avoidance; and a genuine positive CD48 contribution exists only in restricted trans/missing-self contexts. Curators should avoid treating GO:0030101 as a positive mouse function, consider a negative-regulation BP term, and retain GO:0048018 (receptor ligand activity) as the informative molecular function.